NuraFolio Project

NuraFolio helps users create professional portfolios easily using AI and without requiring technical skills.

1. Problem Statement & Objectives

1. **problem statement**

   Many people, such as developers, designers, and students, need to create a portfolio to showcase their skills and projects, but:

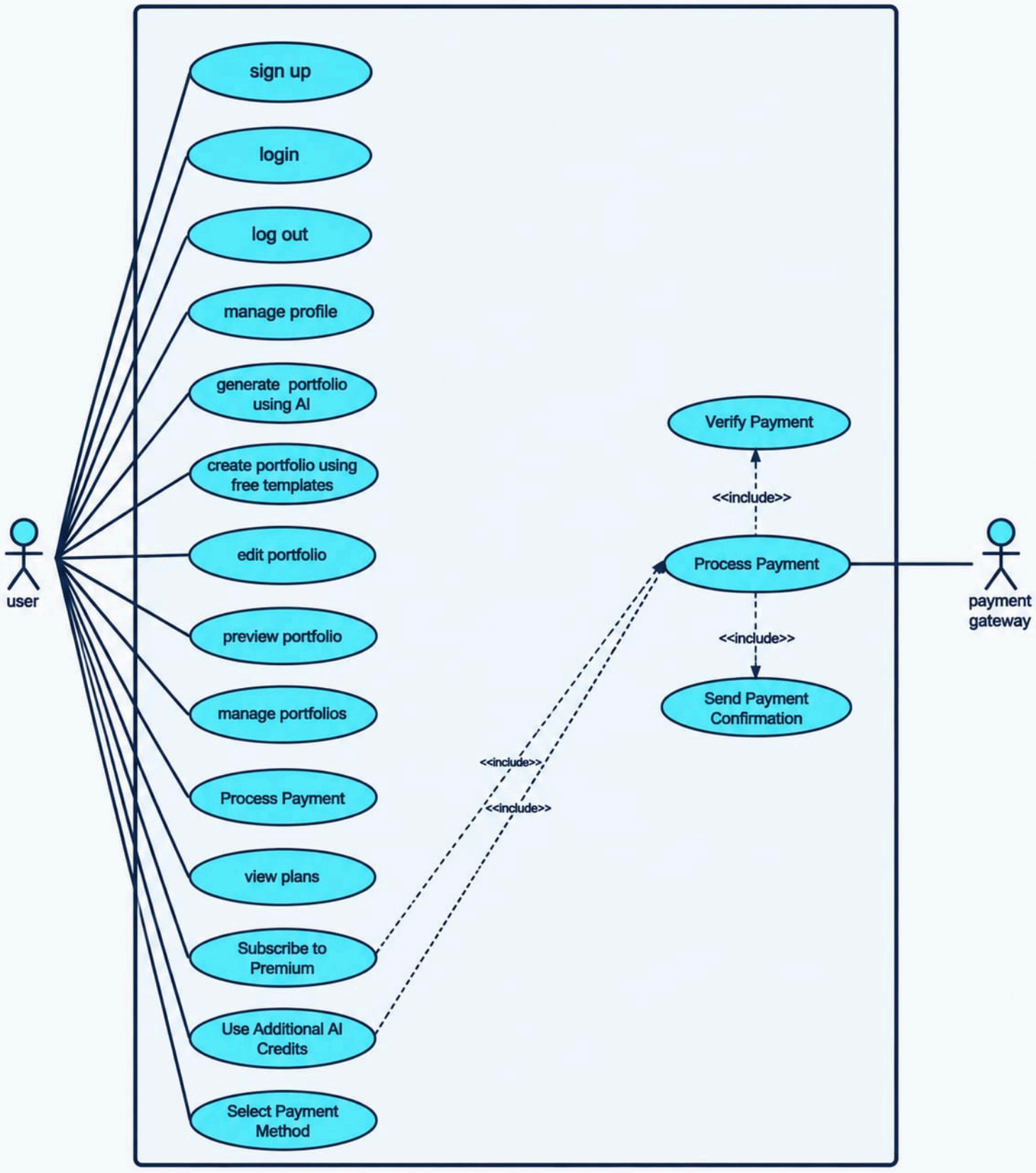
   - Not everyone has experience in website design.
   - Creating a Portfolio from scratch takes time and effort.
   - Some users do not have enough programming experience.
   - Modifying and publishing a website can be complicated for beginners.

2. **objectives**
   - Enable users to create professional portfolios easily.
   - Provide customizable portfolio templates.
   - Use AI to help users generate portfolio content.
   - Allow users to edit and preview their portfolios.
   - Enable users to publish and share their portfolios.

2. Use Case Diagram & Descriptions

1. Use Case Diagram

NuraFolio - AI Portfolio Generator



2. Description

Use Case ID: UC-01

Use Case Name: Sign Up

Primary Actor: User

Description: Allows a new user to create an account on the NuraFolio platform.

Preconditions:

- The user has access to the NuraFolio platform.
- The user does not have an existing account with the same email address.

Trigger: The user selects the Sign Up option.

Main Flow:

1. The user opens the Sign Up page.
2. The system displays the registration form.
3. The user enters the required information.
4. The user submits the registration form.
5. The system validates the entered information.
6. The system creates the user account.
7. The system confirms successful registration.
8. The system redirects the user to the appropriate page.

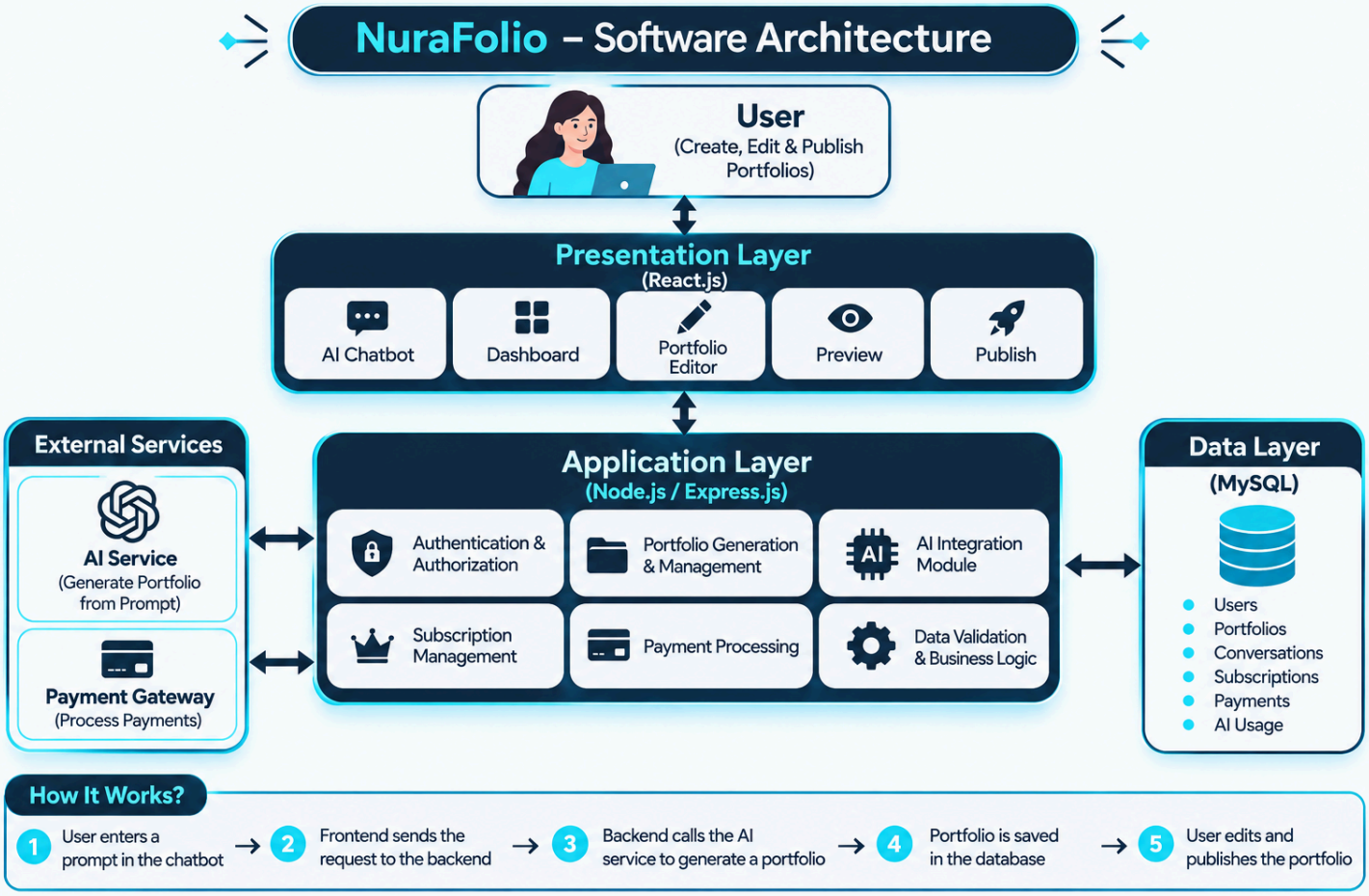
Alternative / Exception Flows:

- If required information is missing or invalid, the system displays validation messages.
- If the email address is already registered, the system informs the user and requests a different email address.

Postconditions:

- A new user account is created successfully.

NuraFolio – Software Architecture



How It Works?

1. User enters a prompt in the chatbot
2. Frontend sends the request to the backend
3. Backend calls the AI service to generate a portfolio
4. Portfolio is saved in the database
5. User edits and publishes the portfolio

Use Case ID: UC-02

Use Case Name: Login

Primary Actor: User

Description: Allows a registered user to access their NuraFolio account.

Preconditions:

- The user has an existing account.
- The user is not currently logged in.

Trigger: The user selects the Login option.

Main Flow:

1. The user opens the Login page.
2. The system displays the login form.
3. The user enters their credentials.
4. The user submits the form.
5. The system validates the credentials.
6. The system authenticates the user.
7. The system redirects the user to the dashboard.

Alternative / Exception Flows:

- If the credentials are incorrect, the system displays an error message.
- If required fields are empty, the system requests the missing information.

Postconditions:

- The user is authenticated and has access to their account.

Use Case ID: UC-03

Use Case Name: Logout

Primary Actor: User

Description: Allows an authenticated user to securely end their current session.

Preconditions:

- **The user is logged in.**

Trigger: The user selects the Logout option.

Main Flow:

1. **The user selects Logout.**
2. **The system terminates the current session.**
3. **The system redirects the user to the Login page or public landing page.**

Alternative / Exception Flows:

- **If the session has already expired, the system redirects the user to the Login page.**

Postconditions:

- **The user is logged** out.

Use Case ID: UC-04

Use Case Name: Manage Profile

Primary Actor: User

Description: Allows the user to view and update their personal account information.

Preconditions:

- The user is logged in.

Trigger: The user opens the Profile section.

Main Flow:

1. The user opens the Profile page.
2. The system displays the current profile information.
3. The user edits the desired information.
4. The user submits the changes.
5. The system validates the updated information.
6. The system saves the changes.
7. The system displays a success message.

Alternative / Exception Flows:

- If the entered information is invalid, the system displays validation messages.
- If saving fails, the system displays an error message.

Postconditions:

- The user's profile is updated successfully.

Use Case ID: UC-04

Use Case Name: Manage Profile

Primary Actor: User

Description: Allows the user to view and update their personal account information.

Preconditions:

- The user is logged in.

Trigger: The user opens the Profile section.

Main Flow:

1. The user opens the Profile page.
2. The system displays the current profile information.
3. The user edits the desired information.
4. The user submits the changes.
5. The system validates the updated information.
6. The system saves the changes.
7. The system displays a success message.

Alternative / Exception Flows:

- If the entered information is invalid, the system displays validation messages.
- If saving fails, the system displays an error message.

Postconditions:

- The user's profile is updated successfully.

Use Case ID: UC-05

Use Case Name: Generate Portfolio with AI

Primary Actor: User

Description: Allows the user to generate a personalized portfolio using AI based on the information provided.

Preconditions:

- The user is logged in.
- The user has the required AI credits or an available generation allowance.

Trigger: The user selects the AI Portfolio Generation option.

Main Flow:

1. The user selects Generate Portfolio with AI.
2. The system displays the required input fields.
3. The user enters their personal, professional, and portfolio-related information.
4. The user submits the information.
5. The system validates the provided information.
6. The system checks the user's available AI credits.
7. The system sends the relevant information to the AI service.
8. The AI service generates the portfolio content.
9. The system displays the generated portfolio.
10. The system saves the portfolio if the user chooses to save it.

Alternative / Exception Flows:

- If required information is missing, the system requests the missing information.
- If the user has insufficient AI credits, the system informs the user and offers the available options for obtaining additional credits.
- If AI generation fails, the system displays an error message and allows the user to retry.

Postconditions:

- AI-generated portfolio content is available to the user.
- The portfolio is saved if the user chooses to save it.

Use Case ID: UC-06

Use Case Name: Create Portfolio Using Free Templates

Primary Actor: User

Description: Allows the user to create a portfolio by selecting and filling in a free portfolio template.

Preconditions:

- The user is logged in.
- Free templates are available.

Trigger: The user selects the option to create a portfolio using free templates.

Main Flow:

1. The user opens the template selection page.
2. The system displays the available free templates.
3. The user selects a template.
4. The system displays the portfolio creation form or editor.
5. The user enters the required information.
6. The system validates the information.
7. The system generates the portfolio based on the selected template.
8. The system saves the portfolio.
9. The system confirms successful creation.

Alternative / Exception Flows:

- If the user selects an unavailable template, the system requests another selection.
- If required information is missing, the system requests the missing information.
- If saving fails, the system displays an error message.

Postconditions:

- A portfolio based on the selected free template is created and saved.

Use Case ID: UC-07

Use Case Name: Edit Portfolio

Primary Actor: User

Description: Allows the user to modify the content and appearance of an existing portfolio.

Preconditions:

- The user is logged in.
- The user has an existing portfolio.

Trigger: The user selects an existing portfolio and chooses Edit.

Main Flow:

1. The user opens the portfolio dashboard.
2. The user selects the portfolio to edit.
3. The system opens the portfolio editor.
4. The system displays the current portfolio content.
5. The user modifies the desired content or appearance.
6. The user saves the changes.
7. The system validates and saves the updated portfolio.
8. The system confirms that the changes have been saved.

Alternative / Exception Flows:

- If the portfolio cannot be found, the system displays an error message.
- If the changes cannot be saved, the system informs the user and allows another attempt.

Postconditions:

- The portfolio is updated with the saved changes.

Use Case ID: UC-08

Use Case Name: Preview Portfolio

Primary Actor: User

Description: Allows the user to preview a portfolio before publishing it or finalizing its content.

Preconditions:

- The user is logged in.
- The user has an existing portfolio.

Trigger: The user selects the Preview option.

Main Flow:

1. The user opens the portfolio editor or dashboard.
2. The user selects a portfolio.
3. The user clicks Preview.
4. The system generates a preview of the portfolio.
5. The system displays the portfolio as it would appear to viewers.
6. The user reviews the portfolio.
7. The user returns to the editor if further changes are required.

Alternative / Exception Flows:

- If the portfolio cannot be loaded, the system displays an error message.
- If the portfolio has not been saved, the system follows the application's configured preview behavior.

Postconditions:

- The portfolio preview is displayed to the user.
- No changes are made unless the user explicitly edits and saves the portfolio.

Use Case ID: UC-09

Use Case Name: Manage Portfolios

Primary Actor: User

Description: Allows the user to view and organize their existing portfolios and perform supported management operations.

Preconditions:

- The user is logged in.

Trigger: The user opens the portfolio dashboard.

Main Flow:

1. The user opens the portfolio dashboard.
2. The system retrieves the user's portfolios.
3. The system displays the available portfolios.
4. The user selects a portfolio or a management action.
5. The system performs the selected supported action.
6. The system updates the portfolio list when necessary.

Alternative / Exception Flows:

- If the user has no portfolios, the system displays an empty state and provides an option to create a portfolio.
- If a selected portfolio cannot be found, the system displays an error message.
- If a management operation fails, the system informs the user.

Postconditions:

- The user's portfolio list reflects the completed management operation.

Use Case ID: UC-10

Use Case Name: Process Payment

Primary Actor: User

Supporting Actor: Payment Gateway

Description: Allows the user to submit a payment for a premium subscription or additional AI credits through an external payment gateway.

Preconditions:

- The user is logged in.
- The user has selected a paid plan or an additional AI credit package.
- The user has selected a supported payment method.

Trigger: The user confirms the payment.

Main Flow:

1. The user confirms the selected purchase.
2. The system prepares the payment request.
3. The system sends the payment request to the Payment Gateway.
4. The Payment Gateway processes the payment.
5. The Payment Gateway verifies the payment.
6. The Payment Gateway sends the payment result or confirmation.
7. The system receives and validates the payment result.
8. The system updates the transaction status.
9. The system activates the purchased subscription or credits if the payment is successful.
10. The system displays the transaction result to the user.

Alternative / Exception Flows:

- If the payment is declined, the system informs the user and allows another attempt.
- If the payment request fails, the system displays an error message.
- If the payment status is pending, the system displays the pending status and updates it when a verified result becomes available.

Postconditions:

- If successful, the transaction is recorded and the purchased service is activated.
- If unsuccessful, the purchase is not activated.

Use Case ID: UC-11

Use Case Name: View Plans

Primary Actor: User

Description: Allows the user to view available subscription plans, their prices, and their features.

Preconditions:

- The user can access the NuraFolio platform.

Trigger: The user selects the View Plans or Pricing option.

Main Flow:

1. The user opens the Plans page.
2. The system retrieves the available subscription plans.
3. The system displays plan names, prices, and features.
4. The user reviews the available plans.
5. The user selects a plan if they wish to proceed with a purchase.

Alternative / Exception Flows:

- If plan information cannot be retrieved, the system displays an error message.

Postconditions:

- The available plans and their details are displayed to the user.

Use Case ID: UC-12

Use Case Name: Subscribe to Premium

Primary Actor: User

Description: Allows the user to purchase a premium subscription to access the platform's premium features.

Preconditions:

- The user is logged in.
- At least one premium plan is available.

Trigger: The user selects a premium plan and chooses Subscribe.

Main Flow:

1. The user opens the available plans.
2. The system displays the premium subscription options.
3. The user selects a premium plan.
4. The system displays the plan details and price.
5. The user confirms the subscription purchase.
6. The system initiates the payment process.
7. The payment is processed and verified through the Payment Gateway.
8. The system receives the payment result.
9. If the payment is successful, the system activates the premium subscription.
10. The system confirms the subscription to the user.

Alternative / Exception Flows:

- If payment is declined, the system informs the user and allows another attempt.
- If the selected plan is unavailable, the system asks the user to choose another plan.
- If payment confirmation is delayed, the system displays the transaction status.

Postconditions:

- If payment succeeds, the user's premium subscription becomes active.
- If payment fails, the premium subscription is not activated.

Use Case ID: UC-12

Use Case Name: Subscribe to Premium

Primary Actor: User

Description: Allows the user to purchase a premium subscription to access the platform's premium features.

Preconditions:

- The user is logged in.
- At least one premium plan is available.

Trigger: The user selects a premium plan and chooses Subscribe.

Main Flow:

1. The user opens the available plans.
2. The system displays the premium subscription options.
3. The user selects a premium plan.
4. The system displays the plan details and price.
5. The user confirms the subscription purchase.
6. The system initiates the payment process.
7. The payment is processed and verified through the Payment Gateway.
8. The system receives the payment result.
9. If the payment is successful, the system activates the premium subscription.
10. The system confirms the subscription to the user.

Alternative / Exception Flows:

- If payment is declined, the system informs the user and allows another attempt.
- If the selected plan is unavailable, the system asks the user to choose another plan.
- If payment confirmation is delayed, the system displays the transaction status.

Postconditions:

- If payment succeeds, the user's premium subscription becomes active.
- If payment fails, the premium subscription is not activated.

Use Case ID: UC-14

Use Case Name: Select Payment Method

Primary Actor: User

Description: Allows the user to select a supported payment method for a premium subscription or an additional AI credit purchase.

Preconditions:

- The user is logged in.
- The user has selected a paid product or subscription.
- At least one payment method is available.

Trigger: The user proceeds to the payment page.

Main Flow:

1. The system displays the available payment methods.
2. The user reviews the available methods.
3. The user selects a payment method.
4. The system validates the selection.
5. The system proceeds to the payment process.

Alternative / Exception Flows:

- If the selected method is unavailable, the system asks the user to select another method.
- If no payment methods are available, the system informs the user that payment cannot currently proceed.

Postconditions:

- A supported payment method is selected for the transaction.
- The system is ready to initiate the payment process.

Use Case ID: UC-15

Use Case Name: Verify Payment

Primary Actor: Payment Gateway

Description: Verifies the payment transaction details and determines whether the payment has been successfully authorized or completed.

Preconditions:

- A payment request has been received.
- The transaction contains the required payment information.

Trigger: The Payment Gateway receives a payment verification request as part of the payment process.

Main Flow:

1. The Payment Gateway receives the transaction details.
2. The Payment Gateway validates the transaction information.
3. The Payment Gateway checks the payment status with the relevant payment provider or processor.
4. The Payment Gateway determines the payment result.
5. The Payment Gateway returns the verification result to the calling process.

Alternative / Exception Flows:

- If the transaction information is invalid, the payment is rejected or marked as failed.
- If the payment provider cannot determine the result, the transaction may remain pending.
- If the payment is declined, the verification result indicates failure.

Postconditions:

- The payment verification result is available to the payment process.

Use Case ID: UC-16

Use Case Name: Send Payment Confirmation

Primary Actor: Payment Gateway

Description: Sends the payment result to NuraFolio so that the system can update the transaction status and take the appropriate action.

Preconditions:

- A payment transaction has been processed.
- A payment result is available.

Trigger: The Payment Gateway completes the relevant payment-processing step and has a result to report.

Main Flow:

1. The Payment Gateway prepares the payment result.
2. The Payment Gateway sends the result to NuraFolio through the configured response or notification mechanism.
3. NuraFolio receives the payment result.
4. NuraFolio validates the result using the appropriate verification mechanism.
5. NuraFolio updates the transaction status.
6. NuraFolio activates the purchased service if the payment is confirmed as successful.

Alternative / Exception Flows:

- If the notification cannot be delivered, the Payment Gateway or NuraFolio follows the configured retry or status-check mechanism.
- If the payment result cannot be validated, NuraFolio does not activate the purchased service until the result is verified.

Postconditions:

- NuraFolio has received the payment result or the transaction remains pending until confirmation can be obtained.

3. functional requirements & non functional requirements

1. Functional Requirements (FR)

A. User Account Management

- FR-01: The system shall allow users to create an account using their personal information and email address.
- FR-02: The system shall allow registered users to log in securely using their credentials.
- FR-03: The system shall allow users to log out of their accounts.
- FR-04: The system shall allow users to reset their passwords through a password recovery process.
- FR-05: The system shall allow users to view and update their profile information.

B. Portfolio Creation and Management

- FR-06: The system shall allow users to create a new portfolio by entering their personal and professional information.
- FR-07: The system shall allow users to create a portfolio by selecting a ready-made template.
- FR-08: The system shall allow users to generate portfolio content using AI based on their provided information or prompts.
- FR-09: The system shall allow users to edit portfolio content, including text, images, and project details.
- FR-10: The system shall allow users to preview their portfolios before publishing them.
- FR-11: The system shall allow users to save their portfolios as drafts.
- FR-12: The system shall allow users to publish and unpublish their portfolios.
- FR-13: The system shall provide each published portfolio with a shareable URL.
- FR-14: The system shall allow users to manage multiple portfolios according to their subscription plan.

C. Template Management

- FR-15: The system shall allow users to browse available portfolio templates.
- FR-16: The system shall allow users to filter templates based on categories or styles.
- FR-17: The system shall allow users to select and apply a template to their portfolios.
- FR-18: The system shall allow users to customize template colors, fonts, and layouts according to their subscription plan.

D. AI Content Generation

- FR-19: The system shall allow users to provide prompts or information to generate portfolio content using AI.
- FR-20: The system shall allow users to regenerate AI-generated content.
- FR-21: The system shall allow users to review and edit AI-generated content before publishing.
- FR-22: The system shall track AI generation usage according to the user's subscription plan.

E. Subscription and Payment Management

- FR-23: The system shall allow users to view available subscription plans and their features.
- FR-24: The system shall allow users to subscribe to a premium plan.
- FR-25: The system shall allow users to select a payment method supported by the payment gateway.
- FR-26: The system shall process subscription payments through an external payment gateway.
- FR-27: The system shall update the user's subscription status after receiving payment confirmation.
- FR-28: The system shall allow users to view their subscription details and payment history.
- FR-29: The system shall restrict premium features based on the user's subscription status.

F. Dashboard and Settings

- FR-30: The system shall provide users with a dashboard to manage their portfolios.
- FR-31: The system shall display portfolio status, such as draft or published.
- FR-32: The system shall allow users to manage account and portfolio settings.
- FR-33: The system shall allow users to delete their portfolios.
- FR-34: The system shall allow users to delete their accounts.

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2. Non-Functional Requirements (NFR)

A. Performance

- NFR-01: The system shall load its main pages within 3 seconds under normal network conditions.
- NFR-02: The system shall display AI generation results within 15 seconds under normal operating conditions, when supported by the AI service.
- NFR-03: The system shall support at least 100 concurrent users without significant performance degradation.

B. Security

- NFR-04: The system shall encrypt sensitive data during transmission using HTTPS.
- NFR-05: The system shall securely store user passwords using a strong hashing algorithm.
- NFR-06: The system shall ensure that users can access and modify only their own private portfolios and account data.
- NFR-07: The system shall validate user inputs to prevent common security vulnerabilities.
- NFR-08: The system shall protect payment information by relying on a secure, compliant external payment gateway.

C. Usability

- NFR-09: The system shall provide a simple and intuitive user interface that can be used by users with basic computer skills.
- NFR-10: The system shall provide clear feedback for successful and unsuccessful user actions.
- NFR-11: The system shall allow users to create and publish a basic portfolio without requiring programming knowledge.

D. Compatibility and Responsiveness

- NFR-12: The system shall support modern web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
- NFR-13: The system shall provide a responsive interface that works on desktops, tablets, and smartphones.

E. Reliability and Availability

- NFR-14: The system shall maintain 99% availability during normal operating conditions.
- NFR-15: The system shall preserve saved portfolio data and prevent data loss during normal operations.
- NFR-16: The system shall handle AI service or payment gateway failures gracefully and inform users of the problem.

F. Maintainability and Scalability

- NFR-17: The system shall use a modular architecture that allows individual components to be updated without affecting unrelated components.
- NFR-18: The system shall support adding new templates and features without requiring major changes to the existing system.
- NFR-19: The system shall be designed to accommodate an increase in users and portfolios as the platform grows.

4.Software Architecture

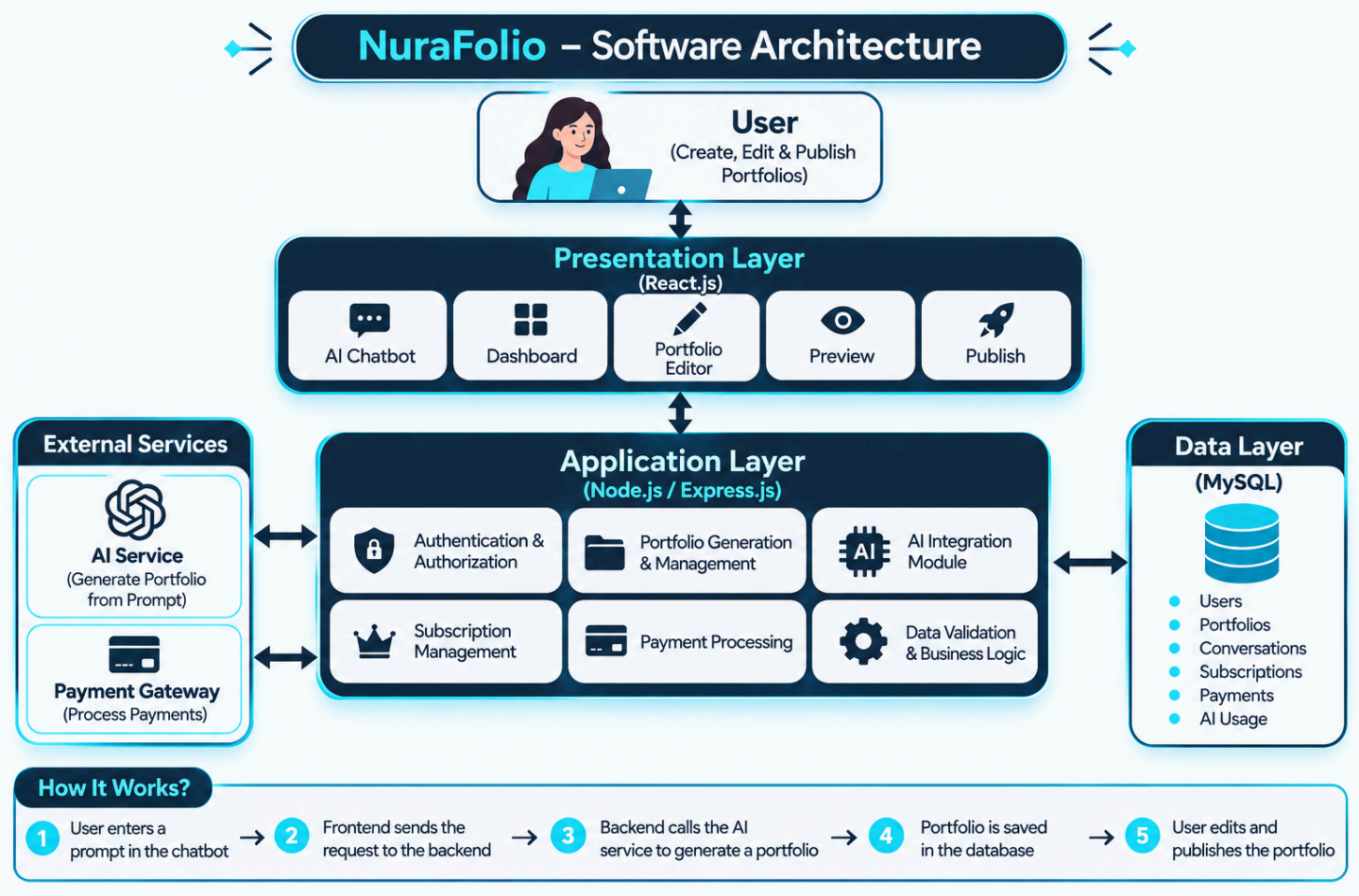
A. Architectural Overview

NuraFolio adopts a Three-Tier Architecture to ensure separation of concerns, maintainability, and scalability. The architecture consists of a Presentation Layer, an Application Layer, and a Data Layer.

The Presentation Layer provides the user interface through which users interact with the system. The Application Layer manages business logic, user requests, and integration with external services, including AI-powered portfolio generation and payment processing. The Data Layer is responsible for storing and retrieving user accounts, portfolio data, conversations, and subscription information.

This architecture enables NuraFolio to generate personalized portfolios from user prompts while maintaining a clear separation between the user interface, business logic, and data management.

B. Architecture Diagram



C. Components Description

Component	Responsibility
Presentation Layer	Provides the chatbot, dashboard, portfolio editor, and user interfaces.
Application Layer	Handles authentication, portfolio generation, editing, publishing, and subscriptions.
Data Layer	Stores and retrieves user, portfolio, conversation, and subscription data.
AI Service	Generates portfolio content and structure based on user prompts.
AI Service	Processes subscription payments and returns transaction results.

D. Data Flow

1. The user submits a prompt through the chatbot.
2. The Presentation Layer sends the request to the Application Layer.
3. The Application Layer validates the request and checks the user's AI credits.
4. The AI Integration Module sends the prompt to the external AI service.
5. The AI service generates the portfolio content and structure.
6. The Application Layer stores the generated portfolio in the database.
7. The Presentation Layer displays the generated portfolio to the user for editing and preview.